



Prevalence, Virulence Gene Profiling, and Characterization of Enteroaggregative *Escherichia coli* from Children with Acute Diarrhea, Asymptomatic Nourished, and Malnourished Children Younger Than 5 Years of Age in India

Vinay Modgil¹, Pankaj Chaudhary², Bhavneet Bharti², Jaspreet Mahindroo¹, Md Yousuf¹, Meenakshi Koundal¹, Balvinder Mohan¹, and Neelam Taneja, MD¹

Objective To study the significance of enteroaggregative *Escherichia coli* (EAEC) as a pathogen causing acute diarrhea and a commensal in healthy nourished and malnourished children younger than five years of age in the Chandigarh region and to address possible traits of EAEC virulence genes, biofilm formation, phylogroups, and antibiotic resistance that would be correlated with diarrhea or carriage.

Study design Stool samples were obtained from children with acute diarrhea (n = 548), as well as nourished (n = 550), and malnourished controls without diarrhea (n = 110). *E coli* isolates were confirmed as EAEC by pCVD432 polymerase chain reaction. Multiplex polymerase chain reactions were used to identify 22 virulence-related genes and phylogeny. Antibiotic susceptibility, adherence, and biofilm-forming potential also were studied.

Results Overall, 16.6% of children were malnourished. EAEC detection was greater among children with acute diarrhea (16%) than nourished (6%) and malnourished nondiarrheal controls (2.7%). We found an association of EAEC infections with age <2 years ($P = .0001$) in the diarrheal group. Adhesive variants adhesion fimbriae IV and adhesion fimbriae II were significantly associated with diarrhea. The *aggR* and *aar* genes showed a positive and negative association with the severity of disease ($P = .0004$ and $P = .0003$). A high degree of multidrug resistance was found (73.8%) in the diarrheal group. Most EAEC strains from the diarrheal group belonged to B2 and D phylogroups, whereas strains from non-diarrheal groups, which belonged to phylogroup B1.

Conclusions EAEC is a significant contributor to childhood diarrhea, its presence as a commensal, and the significance of the association of various virulence factors among the EAEC isolated from diarrheal and non-diarrheal stools. These data reinforce the importance of *aggR* and *aar* as positive and negative regulators and the contribution of AAF/II and AAF/IV fimbria for the pathobiology of EAEC. (*J Pediatr* 2021;234:106-14).

Diarrhea was the eighth-leading cause of mortality in 2016, causing more than 1.6 million deaths worldwide.¹ More than one-quarter (26.93%) of diarrheal deaths occurred in children <5 years of age, and ~90% of deaths caused by diarrhea occurred in South Asia and sub-Saharan Africa.¹ Global mortality among children <5 years of age is an estimated 9.7-10.6 million deaths per year, or 26 000-29 000 children per day, of which 53% are attributed to malnutrition.² In India, diarrhea is the third-most common cause of death, accounting for 13% of deaths among children <5 years of age, and killing an estimated 300 000 children per year.³

In every state of India in 2017, malnutrition was the predominant risk factor for death in children <5 years of age, accounting for 68.2% of total deaths.⁴ Children with lower nutritional status have longer episodes of diarrhea than children with better nutrition.⁵⁻⁷ Diarrheagenic *Escherichia coli* (DEC) has been suggested to occur frequently in young children and is the leading cause of pediatric diarrhea in developing countries.⁸⁻¹⁰ Among the various DEC pathotypes (enteroinvasive *Escherichia coli*, enterotoxigenic *Escherichia coli* [ETEC], enterohaemorrhagic *Escherichia coli*, enteroaggregative *Escherichia coli* [EAEC], enteropathogenic *Escherichia coli* [EPEC], and diffusely adherent *Escherichia coli*), EAEC has been cited as a childhood enteric pathogen worldwide.¹¹

AAF	Adhesion fimbriae	MDR	Multidrug resistance
DEC	Diarrheagenic <i>Escherichia coli</i>	M-PCR	Multiplex polymerase chain reaction
EAEC	Enteropathogenic <i>Escherichia coli</i>	OD	Optical density
EPEC	Enterotoxigenic <i>Escherichia coli</i>	PBS	Phosphate-buffered saline
ETEC	Enterohaemorrhagic <i>Escherichia coli</i>	PCR	Polymerase chain reaction
MAL-ED	Malnutrition and enteric disease study	SPATE	Serine protease autotransporter of Enterobacteriaceae
		VRG	Virulence-related gene

From the Departments of ¹Medical Microbiology and ²Pediatrics Medicine, Postgraduate Institute of Medical Education and Research, Chandigarh, India

Supported by the Indian Council of Medical Research (ICMR) grant to the PGIMER (5/8-1(37) 2012-13 ECDII) and University Grant Commission (UGC) (Sr. No. 20614305077, Ref. No.: 22/06/2014(I)EU-V). The authors declare no conflicts of interest.

0022-3476/\$ - see front matter. © 2021 Elsevier Inc. All rights reserved.
<https://doi.org/10.1016/j.jpeds.2021.03.010>

However, in certain cases, EAEC induces only subclinical enteric infection or gut colonization; studies assessing children from low- and middle-income nations did not find a relationship between EAEC and diarrhea.^{12,13} However, EAEC infection without diarrhea leads to childhood stunting.¹⁴

Several studies have correlated diarrhea with several EAEC virulence-related genes (VRGs), and the high heterogeneity of EAEC strains is a significant contributor.^{15,16} The broad range of VRGs include genes linked to adherence factors, proteases, toxins, metabolic enzymes, and other regulatory factors.^{17,18} EAEC strains colonize the intestinal mucosa via adhesion fimbriae (AAF), composed of at least 4 major antigenic variants (AAF/I-AAF/IV), and are regulated transcriptionally by the AggR activator.^{19,20} AggR also contributes to the expression of genes encoding dispersin (*aap*), *aat* (dispersin translocator), and chromosome cluster (*aai*) encoding the type VI secretion system in the EAEC.²¹ Serine protease autotransporter of Enterobacteriaceae (SPATE) toxins of EAEC have been categorized phylogenetically into 2 classes.¹⁹ Members of class 1 SPATEs are cytotoxic to epithelial and include the plasmid-encoded toxin (Pet) and its homologs, (Sat) secreted autotransporter toxin and (SigA) IgA protease-like homolog. Class 2 are noncytotoxic, and these SPATEs include Pic (Serine protease precursor), a mucinase that facilitates intestinal colonization of EAEC.²⁰ EAEC strains generally enhance mucus secretion from the mucous membranes, trapping the microbes in a bacterium–mucus biofilm.²² The formation of a thick biofilm can be linked to its ability to stimulate chronic diarrhea and persistent colonization.²² In addition to producing a mucous biofilm, many EAEC strains trigger cytotoxicity on the gastrointestinal tract. Typical infection is defined by watery, mucoid, secretory diarrhea with low-grade fever and little-to-no vomiting.²³

E. coli strains are genetically categorized into 4 major phylogenetic groups, A, B1, B2, and D.²⁴ Extraintestinal strains of *E. coli* belong primarily to group B2 and, to a small extent, to group D, whereas the majority of commensal strains belong to group A.²⁵ The most DEC strains assign mostly to the phylogroup D, and the commensal strains to groups A and B1.²⁶ EAEC phylogeny is diverse, and EAEC strains are dispersed in all the phylogroups.²⁵ Phylogenetic groups differ in their genotypic and phenotypic traits, comprising their antimicrobial resistance and virulence genes profiles.²⁷

In the malnutrition and enteric disease study (MAL-ED), asymptomatic EAEC infections were prevalent early in life and were correlated with linear growth shortfalls. Associations of EAEC infection with intestinal inflammation were small in magnitude but recommended a pathway for the growth impact.²⁸ Our main objectives were to describe the significance of EAEC as a pathogen causing acute diarrhea and commensal carriage of EAEC in healthy nourished and malnourished young children in the Chandigarh region and to address possible traits of EAEC VRGs that would be correlated with diarrhea or carriage. We assessed drug sensitivity patterns, their phylogenetic relationship, and the biofilm-forming potential of EAEC from children.

Methods

Ethics Statement

The study was approved by the Postgraduate Institute of Medical Education and Research (PGIMER) Ethics Committee (INT/IEC/2017/173). Written informed consent was obtained from the patient's parent/guardian.

Clinical Groups

The study population comprised of 3 groups. Group 1 included 548 children with acute community-acquired diarrhea aged 0-5 years, group 2 consisted of 550 healthy, well-nourished controls (without diarrhea) aged 0-5 years, and group 3 included 110 malnourished controls (without diarrhea) aged 0-5 years.

Children with Acute Diarrhea

Diarrhea was defined as ≥ 3 liquid or semiliquid stools passed in a 24-hour period lasting < 7 days. The samples were collected during the 2015-2017 period from the primary health center and subcenters catered by Civil Hospital Manimajra and the daycare diarrhea treatment unit of PGIMER, Chandigarh. Only community-acquired diarrhea was included. Children with intellectual disability, physical challenges, serious medical conditions (such as HIV/tuberculosis, etc), or born premature (< 37 weeks of gestation) were excluded.

Clinical data of the diarrheal episodes were collected from each case. Vesikari score was used to assess an EAEC-associated episode of diarrhea.²⁹ Elements of the ranking include the length of diarrhea (in days; score, 0-3 points), the maximum number of stools per day during the episode (score, 1-3 points), the incidence of vomiting (score, 0-1 point), the maximum number of emeses per day during the episode (value, 0-3 points), the presence of fever (score, 0-1 point), the presence of fatigue (score, 0-1 point), and treaties. Fourteen was the highest score.

Nourished and Malnourished Children without Diarrhea

A total of 550 and 110 stool samples were collected from healthy and malnourished children, respectively, < 5 years who were enrolled in the Anganwadi centers of Chandigarh established by the government and various creches of Chandigarh. An Anganwadi center offers primary healthcare to children as a part of the government public healthcare system.³⁰ In Anganwadis, the screening for malnutrition was done in all these children using their anthropometric data, and subsequently, the malnourished children were selected based on the following inclusion and exclusion criteria.

Inclusion Criteria for Malnourished Controls without Diarrhea

Malnourished control without diarrhea included children with mild-to-severe malnutrition with anthropometric measurements < -2 z scores from the median of World Health Organization reference growth charts—regardless of the

types of malnutrition (underweight, stunted, wasted, or a combination of all).

Exclusion Criteria for Nourished and Malnourished Controls without Diarrhea

Children with intellectual disability, serious illnesses (such as HIV/TB, etc), physical challenges, or born premature (<37 weeks of gestation). Any child with an acute episode of diarrhea (<7 days).

Sample Collection and Processing

Stool samples were collected in sterile containers, transferred to Cary Blair media, and transported to the laboratory in the cold chain. They were cultured for the presence of *Vibrio cholerae*, *Aeromonas* spp., *Salmonella* spp., *Shigella* spp., and *E coli* by standard procedures.³¹ To summarize in brief, samples were inoculated onto MacConkey agar, ampicillin blood agar, xylose lysine deoxycholate agar, thiosulfate–citrate–bile salts–sucrose agar, alkaline peptone water, and selenite F broth and incubated at 37°C for 18–24 hours. Organisms were identified by standard biochemical³¹ and matrix-assisted laser desorption/ionization-time of flight, which was performed on a MALDI Microflex LT mass spectrometer (Bruker Daltonik GmbH), and confirmed by serotyping using antisera from Denka-Seiken (Denka-Seiken Co. Ltd., Japan). Up to 3 confirmed *E coli* colonies were selected from each MacConkey agar plate and streaked onto fresh, sterilized nutrient agar and were stored in trypticase soy broth containing 15% glycerol at –80°C.

Extraction of DNA from Confirmed *E coli*

DNA from each confirmed *E coli* isolates was extracted by the heat extraction method.³² In 100 μ L of deionized water, a single colony of biochemically confirmed *E coli* isolates was emulsified and boiled for 5 minutes. The mixture was centrifuged at 10 000 g for 1 minute, and DNA containing supernatant was collected and stored at 4°C until further usage.

Screening of DEC Genes

Multiplex polymerase chain reaction (PCR) for enterotoxins (heat-labile and heat-stable), EPEC (Eae) protein bundle-forming protein (Bfp), Shiga toxins (Stx1, Stx2), VTcom for enterohaemorrhagic *Escherichia coli*, and (CVD432) for EAEC was performed using published primers and protocol as shown in Table I (available at www.jpeds.com).^{33–37} The pCVD432 primer used in this study for the detection of EAEC amplified the 630-bp region from start position 65 to the end position 694 of the CVD432 gene.

PCR Conditions

In single PCR, one pair of forward and reverse primers, and in M-PCRs, 2 or more pairs of primers were used. All PCRs were performed in 20 μ L of final volume containing 0.5 μ L of the template DNA, 1 μ L of DNA polymerase, buffer, 1.5 mM MgCl₂, and 0.2 mM each dNTP, at the final concentration of 1.5 mM. For the final PCR, additional MgCl₂ to the final

concentration of 2 mM and each of the primers to the final concentration of 10 μ M was added.

The PCR conditions for all types of PCRs were as follows: 95°C for 2 minutes, 95°C for 15 seconds, 52°C for 8 seconds, and 10 seconds 72°C for 30 cycles, with a final 2-minute extension at 72°C, and all the PCRs were performed in the thermocycler. Amplified samples were evaluated by 1.5% agarose gel electrophoresis in tris–borate–EDTA buffer and EtBr staining.

Detection of Virulence Factors by PCR

E coli strains confirmed as EAEC were further investigated for virulence genes. PCR for virulence genes, including *aggR*, *aap*, *aataA*, *pet*, *sigA*, *pic*, *sepA*, *sat*, *aaic*, *agg4A*, *aafA*, *astA*, *sepA*, *sat*, *ORF3*, *aggA*, *agg3A*, *aaf C*, *ORF 61*, *eilA*, *capU*, *air*, *espY2*, and *rmoA* was performed separately in a 25- μ L reaction mixture containing 2.5 μ L of 10 $\times$ PCR buffer, 1 mM MgCl₂, 1 mM each dNTP, 0.5 U *Taq* DNA polymerase, and 5 μ L of DNA. Primer sequences, concentration, and annealing temperatures for detecting virulence genes are described in Table II (available at www.jpeds.com).^{38–41}

Phylogenetic Analysis via PCR

The triplex PCR methods employing phylogenetic group-specific primers for 2 genes, *chuA*, *yjaA*, and a cryptic DNA fragment, TspE4C2 was used to determined phylogenetic groups (A, B1, B2, and D) (Table III; available at www.jpeds.com).⁴² The grouping was coupled with a dichotomous decision tree, according to Clermont et al.²⁴

Antibiotic Sensitivity Testing

Antibiotic sensitivity testing was performed with the disc diffusion method for the following antibiotics: cefepime (15 μ g), piperacillin–tazobactam (10 μ g), ertapenem (10 μ g), ampicillin (10 μ g), ciprofloxacin (5 μ g), gentamicin (10 μ g), cotrimoxazole (25 μ g), cefoxitin (30 μ g), amikacin (30 μ g), imipenem (10 μ g), levofloxacin (5 μ g), and ceftriaxone (30 μ g) as per CLSI guidelines.⁴³ Multidrug resistance was described as acquired resistance to 3 or more antimicrobials from different antimicrobial classes tested.

Quantitative Biofilm Assay

A quantitative biofilm assay was performed according to the protocol, as described by Naoko et al.⁴⁴ To evaluate biofilm formation, we inoculated 200 μ L of 0.45% glucose-rich Muller–Hinton broth (HiMedia Laboratories) in 96-well-bottom microtiter polystyrene plates with 5 μ L of an overnight Luria broth culture grown at 37°C with shaking. The sample was incubated at 37°C overnight (18 hours). After incubation, the supernatant was discarded, and each well was washed with phosphate buffer saline (PBS) and stained with 200 μ L of 0.5% crystal violet. After staining, the plate was washed gently to remove the extra stain, and crystal violet was dissolved by adding ethanol, and optical density (OD) was taken at 570 nm in an enzyme-linked immunosorbent assay reader.

EAEC isolates having an OD value >1 were interpreted as strong biofilm producers. Bacteria with an OD value >0.5 < 1 were considered weak biofilm producers and isolates with an OD value <0.4 were noted as nonbiofilm producers.⁴⁴

HEp-2 Cell Adherence Assay

All EAEC isolates were subjected to HEp-2 adherence tests.⁴⁵ HEp-2 cells were grown overnight to 50% confluence in Dulbecco's Modified Eagle's medium containing penicillin, streptomycin, and 2% fetal bovine serum on 8-well chamber slides. Bacteria were grown for 16 hours in Luria broth without shaking. The HEp-2 cells were washed 5 times with PBS, and then the medium was replaced with Dulbecco's Modified Eagle's medium containing 1% mannose. A volume of 10 μ L of bacterial suspension was added per well, and the slides were incubated at 37°C in 5% CO₂ for 3 hours. The monolayers were rinsed 5 times with PBS, then fixed with 70% methanol and Giemsa. The strains were stained with 10% methylene blue dye. Each strain was analyzed in duplicate, and appropriate controls were included in the test. Strains that adhered to the monolayers were documented as adhering in localized, diffuse, or aggregative patterns.

Statistical Analyses

A 2-tailed χ^2 test was used to compare groups. If low predicted values constrained the study, the Fisher exact test was used. OR and 95% CIs were calculated using the GraphPad PRISM version 5.0 software (GraphPad Software, San Diego, CA).

Results

Distribution of DEC Pathotypes Among all Groups

Stool specimens from children with diarrhea (n = 548) and nourished (n = 550) and malnourished (n = 110) children were collected during 2015-2017. DEC pathotypes were detected by M-PCR, as shown in Table IV. The distribution of DEC among various age groups is depicted in Tables IV, V, and VI. The most frequent category of DEC was EAEC (16%), followed by EPEC (7.6%), and ETEC (6.5%) in the acute diarrheal group (Table IV). Among children from the nourished and malnourished control groups, the most prevalent DEC was EAEC (6%) and ETEC (6.36%), respectively, as shown in Tables IV, V, and VI.

Following statistically significant correlations of age and illness with a particular DEC category were observed. EAEC in the acute diarrheal group was found more often in children aged 0-2 years than children aged 2-5 years ($P = .0001$), shown in Table IV. Similarly, a statistically significant correlation between age and prevalence of DEC and EPEC ($P = .0001$ and $P = .01$, respectively) was found (Table IV).

Among 550 children from the nourished control group (without diarrhea), a total of 77 (14%) DEC were detected via M-PCR. The following DEC pathotypes were identified: EAEC 6% (33/77), EPEC 4% (21/77), and ETEC 4.18% (23/77) (Table V). In the nourished control group, there

Table IV. Distribution of DEC groups by age in children with acute diarrhea

DEC groups	Number of children (%)	Child age, y (%)			P value
		0-1	1-2	2-5	
Total DEC	548	80	110	358	
EAEC	82 (30)	34 (42.5)	48 (43.6)	82 (22.9)	.0001*
EPEC	86 (16)	16 (20)	30 (27)	40 (11.7)	.0001*
EPEC	42 (7.6)	10 (12.5)	12 (10.9)	20 (5.5)	.01*
ETEC	36 (6.5)	8 (10)	6 (5.4)	22 (6.1)	.5

*Statistically significant ($P < .05$) when compared pooled prevalence of EAEC in children aged 0-2 years to children aged 2-5 years. The Fisher exact test was used to compare the presence of DEC in children in ADC group at different age groups.

Table V. Distribution of DEC groups by age in nourished controls without diarrhea

DEC groups	Number of children (%)	Child age, y (%)			P value
		0-1	1-2	2-5	
Total DEC	n = 550	n = 109 (19.8%)	n = 149 (27%)	n = 292 (53%)	
EAEC	77 (14%)	21 (19.2%)	25 (32.4%)	31 (10.6%)	.001*
EPEC	33 (6.0%)	9 (8.2%)	10 (6.7%)	14 (4.79%)	.2
EPEC	21 (4.0%)	5 (4.5%)	10 (6.7%)	6 (2.05%)	.02*
ETEC	23 (4.18%)	7 (6.4%)	6 (4.02%)	10 (3.42%)	.3

*Statistically significant ($P < .05$) when compared pooled prevalence of EAEC in children aged 0-2 years to children aged 2-5 years. The Fisher exact test was used to compare the presence of DEC in children in nourished controls without diarrhea at different age groups.

were statistically significant associations between age and infection with a specific DEC group as follows; DEC infection was more frequently observed among children belonging to 0-2 years age group than children in the 2-5 years of age group ($P = .001$) shown in Table V. Also, there were statistically significant associations of EPEC infection with age group 0-2 vs 2-5 years of age ($P = .02$). Overall, DEC and EPEC in healthy children were significantly associated in children 0-2 years. However, in the malnourished control group, there was no significant difference ($P = 1.0$) observed, as shown in Table VI.

The prevalence of DEC was observed to be 30% in the acute diarrheal group of children, whereas in the nourished and malnourished control groups, it was 14%. Among DEC pathotypes, EAEC was the most prevalent pathotype, accounting for 16% in the acute diarrheal group and 5.4% in the nourished and malnourished control groups, as shown in Table VII (available at www.jpeds.com). The overall prevalence of DEC was statistically significant when compared with the control or healthy individuals ($P = .0001$) (Table VII). Also, the occurrence of EAEC and EPEC was statistically significant among the 2 groups (Table VII).

Severity in Acute Diarrhea Group and Children Infected with EAEC Based on the Number, Age, and Sex

The severity score of the stool samples collected from the Chandigarh region from children with acute diarrhea. Most

Table VI. Distribution of DEC groups by age in malnourished controls without diarrhea

DEC groups	Number of children (%) n = 110	Child age, y (%)			P value
		0-1	1-2	2-5	
		n = 25 (22.7%)	n = 30 (27.2%)	n = 55 (50%)	
Total DEC	15 (13.6%)	6 (24%)	4 (13.3%)	5 (9.09%)	.2
EAEC	3 (2.7%)	0 (0)	2 (6.6%)	1 (1.8%)	1.0
EPEC	5 (4.5%)	3 (12%)	0 (0)	2 (3.6%)	1.0
ETEC	7 (6.3%)	3 (12%)	2 (6.6%)	2 (3.6%)	.4

There is no statistical significant difference found when the pooled prevalence of EAEC in children aged 0-1, 1-2 years to children aged 2-5 years. The Fisher exact test was used to compare the presence of DEC in children in malnourished controls without diarrhea at different age groups.

of the patients presented with severe diarrhea (42.3%) and moderate diarrhea (25.8%). When severe and moderate severity was pooled and compared with mild severity, a statistically significant difference among children with diarrhea was observed ($P < .0001$).

The severity score of cases from whom EAEC was isolated is shown in **Table VIII** (available at www.jpeds.com). Severity of EAEC among children with acute diarrhea was as follows; severe 15 (27.7%); mild 34 (40.9%); and moderate 26 (31.3%). Correlation of severity with age found that children belonging to 0-2 years of age group were more (55.4%) affected in comparison with the children 2-5 years of age (48%) (**Table VIII**).

A total of 110 children aged <0-5 years were included in the study. The severity of malnutrition was reported as follows; stunting (57.2%) was the most common nutritional abnormality observed in this study, followed by wasting (46.7%), and underweight (40.9%) (**Table IX**; available at www.jpeds.com).

Distribution of VRGs of EAEC Among all Groups

To evaluate the roles of putative virulence genes in EAEC epidemiology, 4 M-PCR assays were used to describe 21 genes previously found in EAEC strains (**Table X**). Of the 21 genes scored, *astA* was the most frequently detected (91%), followed by *aggR* (83.7%) and *aap* (58.1%) among children from the acute diarrheal group (**Table X**). The *aggR*-positive EAEC isolates were mostly detected in children with diarrhea (83.7%) as compared with nondiarrheal (36.1%) controls. Of the 5 genes encoding SPATEs, the most common were *sat* (25.5%), *pic* (16.2%), and *sepA* (6.9%), and the least common SPATEs was *pet* (4.6%) among the acute diarrheal group. Among adhesive genes, AAF/IV was the most common (34.8%) in the EAEC isolated from diarrheal stools, and it was present in 13.8% of EAEC isolated from nondiarrheal stools from nourished and malnourished children. The ORF61, which is encoded by *aar* gene (repressor of *aggR* gene), was more prevalent in EAEC from the nondiarrheal stools (69.4%) compared with diarrheal stools (23.2%). Statistically, there was a positive association in ORF3 ($P = .006$), *aggR* ($P = .004$), *agg4A* ($P = .04$), and *aafC* ($P = .0002$) genes and negative association in *pet*

Table X. Distribution of VRGs of EAEC among children in all groups

Virulence genes	Group 1 acute diarrheal, n = 86 (%)	Group 2 nourished n = 33 (%)	Group 3 malnourished n = 3 (%)	Total n = 122 (%)	P value
<i>astA</i>	80 (91)	30 (90.9)	3 (100)	113 (92.6)	.7
<i>sigA</i>	4 (4.6)	2 (6.0)	0 (0)	6 (5)	1.0
<i>pic</i>	14 (16.2)	3 (9.0)	1 (33.3)	18 (14.7)	.7
<i>sepA</i>	6 (6.9)	3 (9.0)	1 (33.3)	10 (8.1)	.6
<i>sat</i>	22 (25.5)	3 (9.0)	1 (33.3)	26 (21.3)	.1
<i>pet</i>	4 (4.6)	10 (30.3)	1 (33.3)	15 (12.2)	.0002*
ORF3	22 (25.5)	18 (54.5)	2 (66.6)	42 (34.4)	.003*
<i>aap</i>	50 (58.1)	16 (48.4)	3 (100)	69 (56.5)	.8
<i>aafC</i>	14 (16.2)	8 (24.2)	1 (33.3)	23 (18.8)	.4
<i>aggR</i>	72 (83.7)	11 (33)	2 (66.6)	85 (69.6)	.0004*
<i>agg4A</i>	30 (34.8)	3 (9.0)	2 (66.6)	35 (28.6)	.04*
<i>aggA</i>	18 (20.9)	4 (12.1)	1 (33.3)	23 (18.8)	.5
<i>aafA</i>	6 (6.9)	2 (6.0)	0 (0)	8 (6.5)	1.0
<i>agg3A</i>	10 (11.6)	1 (3.0)	0 (0)	11 (9)	.2
<i>aafC</i>	32 (37.2)	1 (3.0)	0 (0)	33 (27)	.0002*
<i>aar</i>	20 (23.2)	22 (66.6)	3 (100)	49 (40.1)	.0003*
<i>eilA</i>	38 (39.5)	15 (45.4)	1 (33.3)	54 (44.2)	.6
<i>capU</i>	32 (37.2)	15 (45.4)	2 (66.6)	49 (48)	.3
<i>espy</i>	24 (27.9)	4 (12.12)	1 (33.3)	29 (23.7)	.1
<i>rmoA</i>	44 (51)	15 (45.4)	2 (66.6)	61 (50)	.8
<i>shlA</i>	32 (37.2)	9 (27.2)	0 (0)	41 (33.6)	.3
<i>air</i>	22 (25.5)	4 (12.12)	0 (0)	26 (21.3)	.1

*Statistically significant ($P < .05$) when EAEC strains from the acute diarrheal group were compared with nourished and malnourished control groups. Data was analyzed by using the Fisher exact test.

($P = .02$) and *aar* (0.0003) genes with diarrhea when EAEC strains from the acute diarrheal group were compared with nourished and malnourished control groups (**Table X**).

Based on virulence genes analysis in EAEC isolates among diarrheal and control samples, several different combinations of virulence markers were found among the EAEC isolates in our study. The gene combination *astA*, *aggR*, *agg4A* (encoding AAF/IV) was the most prominent (11.6%) among diarrheal stool isolates. This gene combination was present in 34% of EAEC isolates from the diarrheal group in combination with other genes (**Table XI**; available at www.jpeds.com). This gene combination was detected only in 6 EAEC (16%) isolates from nondiarrheal stools. This combination was found statistically significant ($P = .05$) when diarrheal and nondiarrheal strains were compared. We conclude that the presence of *astA*, *aggR*, *agg4A* would co-relate with the pathogenicity of EAEC. The presence of *aar* gene encoded by ORF61, present in 69.4% in EAEC isolated from nondiarrheal stools, may signify less virulence of EAEC isolates, as shown by our data (**Table XI**). In our findings, 41 (47.6%) of EAEC strains were lacking *aar* and AAF/IV in the diarrheal group whereas in the nondiarrheal group, *aar* and AAF/IV were absent in 10 (27.7%) of EAEC strains. The absence of *aar* and AAF/IV was statistically significant ($P = .04$) when the diarrheal and nondiarrheal groups were compared.

Among EAEC isolates, the phylogenetic group (D [37.2%], B1 [39.3%], and B1 [66.6%]) were the most prevalent in children with acute diarrhea and nourished and malnourished nondiarrheal controls groups, respectively. There were no

statistically significant differences in phylogroups when EAEC strains from the acute diarrheal group were compared with nourished and malnourished controls groups (Table XII; available at www.jpeds.com).

Biofilm Analysis, Antibiotic Resistance, and Adherence of EAEC Strains

Biofilm analysis revealed that 27.4% of EAEC isolates from the acute diarrheal group were strong, and 53.2% of isolates were weak biofilm producers, whereas 19.3% were nonbiofilm producers. Similarly, among the nourished nondiarrheal control group, 42.4% of EAEC isolates were strong biofilm producers, and in malnourished children, all the EAEC strains were weak biofilm producers.

All strains were found to be resistant to at least 1 of the antibiotics tested. The greatest antibiotic resistance was observed in ampicillin (87%), ciprofloxacin (72.5%), cotrimoxazole (72.5%), ceftriaxone (66.1%), and levofloxacin (54.8%) in EAEC isolates from the acute diarrheal group (Figure 1; available at www.jpeds.com). Antibiotic resistance was greater in the acute diarrheal group than the EAEC isolates from nourished and malnourished nondiarrheal controls groups. Multidrug resistance (MDR), defined as acquired resistance toward ≥ 3 antibiotics from different antibiotic classes tested, was observed in 73.8% of the EAEC strains.

Aggregative adherence patterns of EAEC identified by the HEp-2 adhesion test are shown in Figure 2 (available at www.jpeds.com). Three aggregative adherence types were observed in the present study: type A, the typical aggregative adherence pattern with a honeycomb formation; type B, an aggregative adherence pattern without the typical honeycomb formation; and type C, an aggregative adherence pattern with lined up cells. The prevalence of each aggregative adherence pattern in EAEC strains are shown in Figure 2, A-D. Most strains showed the typical aggregative adherence pattern (type A), as shown in Figure 3 (available at www.jpeds.com).

Discussion

Malnutrition is the predominant risk factor for death in children <5 years of age in India, accounting for 68.2% of deaths and the leading risk factor for health loss for all ages.⁴ The present study was carried out in different Anganwadi centers of urban areas of Chandigarh, which is a union territory that serves as the capital of the 2 neighboring states of Punjab and Haryana. In our study, overall, 16.6% of children aged 0-5 years were found to be malnourished, where 9.5% were stunted (height for age), 8% were wasted (weight-for-height), and 6.8% were underweight. In the Food and Nutrition Security survey, 17.8% of children of 0-5 years of age were stunted and underweight children, 10.3% were wasted and underweight, and 6.1% were stunted, wasted, and underweight. In Anganwadi, a mid-day meal is distributed as part of the Integrated Child Development Services program to combat child hunger and malnutrition.

According to the MAL-ED study, EAEC infection can affect the development of children by causing intestinal inflammation and impairing the growth of children, even in the absence of diarrhea.²⁸ In MAL-ED, the investigators used molecular methods to score the abundance of the pathogen and tested multiple specimens over time. Point prevalence using culture (much less sensitive) probably explains the lack of association, especially because asymptomatic carriage usually occurs at lower abundance. In MAL-ED study, EAEC infections were common in asymptomatic children at an early age of life, along with clinical growth shortfalls, and the greatest impact of EAEC was observed among children with the greatest frequency of detection during the first 2 years of life.²⁸ The presence of EAEC without diarrhea also has been associated with a decreased height-to-age score and chronic inflammation in Brazilian children.¹² Despite a large number of reported EAEC VRGs and their detailed studies on diarrhea-related strains, there are few data on the association of EAEC infection with different clinical outcomes, such as malnutrition.¹² The high heterogeneity of EAEC strains is a major contributor in this contexts.³⁴

In this study, the EAEC detection was based on PCR with primers complementary for pCVD432, an assay representing a reliable method of detecting EAEC.³⁵ We found a high prevalence of EAEC among the acute diarrheal group (16%), corroborated by other studies that employed this virulence marker for EAEC detection.^{36,37,39,40} A study conducted at NICED, Kolkata, India, reported a 12% prevalence in children <5 years of age.⁴¹ However, studies from various parts of India reported variable prevalence (7.0%-16.0%) of EAEC.^{42,46} Employing the same target pCVD432, geographic variations are reported. Studies in Britain, Germany, the US, and Romania found EAEC prevalence in diarrhea varying from 2% to 6%.^{36,37,39,40} In Nepal, Brazil, and Mali, the rate of EAEC among diarrhea cases ranged from 4.5% to 39%.¹⁷⁻⁴⁷

The HEp-2 adherence assay is considered to be the “gold standard” for EAEC detection.⁴⁸ In our study, >80% of the EAEC strains from the pediatric population show an aggregative adherence pattern on HEp-2 cell lines. A study conducted in Germany identified 86% of EAEC strains positive by the HEp-2 cell adhesion assay.⁴⁹

EAEC also is isolated frequently from nondiarrheal stools of nourished and malnourished children, present as commensals.⁴⁸ In our study, we found that the prevalence of EAEC among children of nourished and malnourished nondiarrheal controls group was 6% and 2.7%, respectively. These results were in accordance with a study from Dhaka, Bangladesh, and Manipal, South India, where a high rate of EAEC as commensals (7%) and 9%, respectively, was reported in young children.^{42,50} In Mali, almost 50% (61/121) of EAEC were isolated from the children aged 0-59 months without diarrhea.³⁸ However, a lower prevalence is recorded in developed countries, eg, a study from the US reported a lower prevalence of EAEC (1.7% and 2%) from asymptomatic volunteers.^{51,52} Although in vivo

experiments also have shown that malnutrition increases the susceptibility to EAEC colonization.⁵³ we did not find such a correlation in our study. The probable explanation for this is that better supplementary nutrition, nonformal preschool education, health check-up, vaccination, and health education are provided in Anganwadi centers under the Integrated Child Development Services scheme. The precise mechanism underlying malnutrition linked with EAEC remains unknown but impeded gut microbiota function, and intestinal and systemic inflammation might be associated.

In our data, we found an association of EAEC infections with age ($P = .0001$) in the acute diarrheal group. We found that 0- to 2-year-old children (43%) are highly susceptible to EAEC infection. Children <6 months might be protected from EAEC infection by breastfeeding and be less exposed to bacterial pathogens because most have a limited introduction to solid foods.⁵⁴ The lower prevalence of EAEC among greater age groups is likely due to the development of age-related immunity.⁵⁵ However, due to its heterogeneous nature, the distribution of EAEC infection among children of different age groups varies by geographical region. In our finding, infected children from the acute diarrheal group had moderate to severe forms of diarrhea. Clinical assessment of diarrheal illness showed 48.1% of patients had a severe illness, and 30% of patients presented with a moderate illness where males predominated over females. Our results were in accordance with Ochoa et al, in which 39.1% of severe and moderate diarrheal episodes were due to EAEC infection.⁵⁶

In agreement with previous reports, our samples harbored a diverse range and combination of VRGs.^{38,57} We found a high prevalence of virulence markers of EAEC (including *aap*, *aggR*, adhesive variants) in the acute diarrheal group. AAFs constitute a significant component of EAEC pathogenesis, helping the microorganism to colonize the intestine.⁵⁸ In EAEC, these AAFs include at least 5 major antigenic variants (AAF/I-AAF/V).⁵⁸ A greater prevalence of *agg4A* (encoding AAF/IV fimbrial unit) (34.8%) was observed in the acute diarrheal group, similar to the Brazilian study (38.4%).¹² The AAF/II (*aafA* gene encoding fimbrial subunit) had a small prevalence (6.9%) in the acute diarrheal group, in agreement with other studies investigating this fimbrial subunit.^{38,59} Apart from the *agg4A* gene, *aafC* (encoding the ushers of AAF/II assembly) was also statistically significant (37.2%, P value .002) when the children in the acute diarrheal group were compared with nourished and malnourished non-diarrheal controls group. These data indicate a significant role of AAF/IV and AAF/II for EAEC infection in this population, suggesting an association with more severe infection. AAF/II is not commonly associated with diarrhea.^{38,60} except for the studies performed in Nigeria and Brazil found that the *aafA* encoding AAF/II fimbrial subunits was associated with diarrhea.^{61,62} A genomic analysis, that includes a description of AAF/V, detected in 16% of EAEC strains; in our study, adhesive variant AAF/V was not interrogated.⁶³ The observed differences between fimbrial genes and their respective fimbrial usher genes imply a genetic difference of

these regions inside the subtypes of fimbriae. These results indicate that most EAEC strains possess an AAF adhesin.

The *aggR* (master regulator gene), an EAEC transcription regulator, regulates the expression of several virulence factors. Typical and atypical EAEC were defined as strains with or without *aggR* genes.⁶⁴ The *aggR* gene was increasingly detected from the acute diarrheal group ($P = .0006$). Also, EAEC isolates from the acute diarrheal group harbored more virulence genes, including chromosomal (*aaiC*, *air*, *eilA*, and *pic*) and plasmid genes (*aap*, ORF3, ORF61, *capU*, *sat*, *pet*, and *astA*) than EAEC isolates from nourished and malnourished nondiarrheal controls groups. The *aggR* negative EAEC isolates were more widespread in the nourished nondiarrheal controls group (66%) than the acute diarrheal group (27.5%). We also found a higher prevalence of the *aar* gene in EAEC isolates from malnourished and nourished nondiarrheal controls groups ($P = .0003$), which is a negative regulator of *aggR*.⁶⁵ These results are similar to previously reported case-control studies, where the prevalence of *aar* gene was high in nondiarrheal EAEC isolates.^{38,59} We believe that the combination of *astA*, *aggR*, *agg4A* present in 34% of acute diarrheal ($P .05$) isolates and would correlate with the pathogenicity of EAEC.

We detected a high antibiotic resistance among the EAEC strains. Several reports have documented a high degree of multidrug resistance in EAEC strains.^{66,67} In our study, 73.8% of EAEC strains from the acute diarrheal group were MDR. EAEC isolates from the acute diarrheal group showed higher antibiotic resistance (ampicillin, ciprofloxacin, levofloxacin, cotrimoxazole, and ceftriaxone) than EAEC isolates from nourished and malnourished non-diarrheal controls groups. The greater antibiotic resistance in EAEC may be due to the widespread use of antibiotics.⁶⁸ As EAEC has become increasingly resistant to different antibiotics, the selection of suitable antibiotics must take into consideration in parts of the world where the infection was acquired, as there are different patterns of antimicrobial susceptibility for each geographic region.^{67,69} Resistance to ciprofloxacin was 72.5% in the diarrheal group. We found a lower antibiotic resistance to ertapenem (3.2%) in the acute diarrheal group. In severe cases, ertapenem and azithromycin can be used.

Biofilm formation is considered a pivotal step in developing an infection in human mucosa and cause of MDR. The ability to adhere to different surfaces and the formation of biofilms has been regarded as a vital feature of *E coli* pathogenicity in EAEC. Surface virulence factors of the bacterium, including various adherence factors, can contribute to bacterial adhesion and biofilm growth. Overall, among EAEC isolates from the acute diarrheal group, 80.6% of isolates were able to form a biofilm. Biofilm-forming potential was greater in the acute diarrheal group than the malnourished and nourished nondiarrheal controls group.

The phylogenetic analysis revealed that EAEC phylogeny is diverse and dispersed in all the phylogroups, similar to previous studies.^{70,71} From the acute diarrheal group, most EAEC strains belonged to phylogroup B2 and D, showing

their ability to cause extraintestinal infection. In contrast, EAEC strains from the nourished and malnourished non-diarrheal controls belonged to phylogroup B1 showing their commensal nature. Our findings are in accordance with the study conducted in Mali, which revealed the diverse and dispersed nature of phylogeny among EAEC isolates.¹⁷

This study is limited by a smaller number of samples taken from malnourished children. Our data support EAEC as a significant contributor to childhood diarrhea, its presence as a commensal, and the significance of the association of various virulence factors among the EAEC isolated from diarrheal and nondiarrheal stools. ■

We acknowledge the kind support of local health authorities and Anganwadi health workers for facilitating the sample access in their respective areas.

Submitted for publication Nov 28, 2020; last revision received Mar 4, 2021; accepted Mar 5, 2021.

Reprint requests: Neelam Taneja, Department of Medical Microbiology, Postgraduate Institute of Medical Education and Research Chandigarh, Chandigarh 160012, India. E-mail: drneelampgi@yahoo.com

References

- Naghavi M, Abajobir AA, Abbafati C, Abbas KM, Abd-Allah F, Abera SF, et al. Global, regional, and national age-sex specific mortality for 264 causes of death, 1980-2016: a systematic analysis for the Global Burden of Disease Study 2016. *Lancet* 2017;390:1151-210.
- Guerrant RL, Oriá RB, Moore SR, Oriá MOB, Lima AAAM. Malnutrition as an enteric infectious disease with long-term effects on child development. *Nutr Rev* 2008;66:487-505.
- Million Death Study Collaborators Bassani DG, Kumar R, Awasthi S, Morris SK, Paul VK, Shet A, et al. Causes of neonatal and child mortality in India: a nationally representative mortality survey. *Lancet* 2010;376:1853-60.
- Swaminathan S, Hemalatha R, Pandey A, Kassebaum NJ, Laxmaiah A, Longvah T, et al. The burden of child and maternal malnutrition and trends in its indicators in the states of India: the Global Burden of Disease Study 1990-2017. *Lancet Child Adolesc Health* 2019;3:855-70.
- Black RE, Brown KH, Becker S. Malnutrition is a determining factor in diarrheal duration, but not incidence, among young children in a longitudinal study in rural Bangladesh. *Am J Clin Nutr* 1984;39:87-94.
- James JW. Longitudinal study of the morbidity of diarrheal and respiratory infections in malnourished children. *Am J Clin Nutr* 1972;25:690-4.
- Bairagi R, Chowdhury MK, Kim YJ, Curlin GT, Gray RH. The association between malnutrition and diarrhoea in rural Bangladesh. *Int J Epidemiol* 1987;16:477-81.
- Canizalez-Roman A, Flores-Villaseñor HM, Gonzalez-Nuñez E, Velazquez-Roman J, Vidal JE, Muro-Amador S, et al. Surveillance of diarrheagenic *Escherichia coli* strains isolated from diarrhea cases from children, adults and elderly at Northwest of Mexico. *Front Microbiol* 2016;7.
- Jafari F, Garcia-Gil LJ, Salmanzadeh-Ahrabi S, Shokrzadeh L, Aslani MM, Pourhoseingholi MA, et al. Diagnosis and prevalence of enteropathogenic bacteria in children less than 5 years of age with acute diarrhea in Tehran children's hospitals. *J Infect* 2009;58:21-7.
- Saeed A, Abd H, Sandstrom G. Microbial aetiology of acute diarrhoea in children under five years of age in Khartoum, Sudan. *J Med Microbiol* 2015;64:432-7.
- Nataro JP. Enteropathogenic *Escherichia coli* pathogenesis. *Curr Opin Gastroenterol* 2005;21:4-8.
- Havt A, Lima IF, Medeiros PH, Clementino MA, Santos AK, Amaral MS, et al. Prevalence and virulence gene profiling of enteropathogenic *Escherichia coli* in malnourished and nourished Brazilian children. *Diagn Microbiol Infect Dis* 2017;89:98-105.
- Kotloff KL, Nataro JP, Blackwelder WC, Nasrin D, Farag TH, Panchalingam S, et al. Burden and aetiology of diarrhoeal disease in infants and young children in developing countries (the Global Enteric Multicenter Study, GEMS): a prospective, case-control study. *Lancet* 2013;382:209-22.
- Steiner TS, Lima AAM, Nataro JP, Guerrant RL. Enteropathogenic *Escherichia coli* produce intestinal inflammation and growth impairment and cause interleukin-8 release from intestinal epithelial cells. *J Infect Dis* 1998;177:88-96.
- Bafandeh S, Haghi F, Zeighami H. Prevalence and virulence characteristics of enteropathogenic *Escherichia coli* in a case-control study among patients from Iran. *J Med Microbiol* 2015;64:519-24.
- Lima AAM, Oriá RB, Soares AM, Filho JQ, De Sousa F, Abreu CB, et al. Geography, population, demography, socioeconomic, anthropometry, and environmental status in the MAL-ED cohort and case-control study sites in Fortaleza, Ceará, Brazil. *Clin Infect Dis* 2014;59:S287-S294.
- Boisen N, Scheutz F, Rasko DA, Redman JC, Persson S, Simon J, et al. Genomic characterization of enteropathogenic *Escherichia coli* from children in Mali. *J Infect Dis* 2012;205:431-44.
- Estrada-García T, Navarro-García F. Enteropathogenic *Escherichia coli* pathotype: a genetically heterogeneous emerging foodborne enteropathogen. *FEMS Immunol Med Microbiol* 2012;66:281-98.
- Navarro-García F, Elias WP. Autotransporters and virulence of enteropathogenic *E. coli*. *Gut Microbes* 2011;2:13-24.
- Navarro-García F, Gutierrez-Jimenez J, Garcia-Tovar C, Castro LA, Salazar-Gonzalez H, Pic Cordova V. An autotransporter protein secreted by different pathogens in the Enterobacteriaceae family, is a potent mucus secretagogue. *Infect Immun* 2010;78:4101-9.
- Dudley EG, Thomson NR, Parkhill J, Morin NP, Nataro JP. Proteomic and microarray characterization of the AggR regulon identifies a pheU pathogenicity island in enteropathogenic *Escherichia coli*. *Mol Microbiol* 2006;61:1267-82.
- Wani SA, Hussain I, Rather MA, Kabli ZA, Nagamani K, Nishikawa Y, et al. Putative virulence genes and biofilm production among typical enteropathogenic *Escherichia coli* isolates from diarrhoeic children in Kashmir and Andhra Pradesh. *Indian J Microbiol* 2012;52:587-92.
- Hebbelstrup Jensen B, Olsen KEP, Struve C, Krogfelt KA, Petersen AM. Epidemiology and clinical manifestations of enteropathogenic *Escherichia coli*. *Clin Microbiol Rev* 2014;27:614-30.
- Clermont O, Bonacorsi S, Bingen E. Rapid and simple determination of the *Escherichia coli* phylogenetic group. *Appl Environ Microbiol* 2000;66:4555-8.
- Jaureguiy F, Landraud L, Passet V, Diancourt L, Frapy E, Guigon G, et al. Phylogenetic and genomic diversity of human bacteremic *Escherichia coli* strains. *BMC Genomics* 2008;26:560.
- Nataro JP, Kaper JB. Diarrheagenic *Escherichia coli*. *Clin Microbiol Rev* 1998;11:142-201.
- Chaudhuri RR, Henderson IR. The evolution of the *Escherichia coli* phylogeny. *Infect Genet Evol* 2012;12:214-26.
- Rogawski ET, Guerrant RL, Havt A, Lima IFN, Medeiros PHQS, Seidman JC, et al. Epidemiology of enteropathogenic *Escherichia coli* infections and associated outcomes in the MAL-ED birth cohort. *PLoS Negl Trop Dis* 2017;11:e0005798.
- Schnadower D, Tarr PI, Gorelick MH, O'Connell K, Roskind CG, Powell EC, et al. Validation of the modified Vesikari score in children with gastroenteritis in 5 US emergency departments. *J Pediatr Gastroenterol Nutr* 2013;57:514-9.
- Kapil U, Pradhan R. Integrated Child Development Services scheme (ICDS) and its impact on nutritional status of children in India and recent initiatives. *Indian J Public Health* 1999;43:21-5.
- Koneman E, Giovanniello O, Klajn D, Preciado M. *Diagnostico microbiológico*. 6th edition. Buenos Aires: Editorial Médica Panamericana; 2008.
- Dilhari A, Sampath A, Gunasekara C, Fernando N, Weerasekara D, Sissons C, et al. Evaluation of the impact of six different DNA extraction

- methods for the representation of the microbial community associated with human chronic wound infections using a gel-based DNA profiling method. *AMB Express* 2017;7:179.
33. Tobias J, Vutukuru SR. Simple and rapid multiplex PCR for identification of the main human diarrheagenic *Escherichia coli*. *Microbiol Res* 2012;167:564-70.
 34. Chattaway MA, Jenkins C, Rajendram D, Cravioto A, Talukder KA, Dallman T, et al. Enteroaggregative *Escherichia coli* have evolved independently as distinct complexes within the *E. coli* population with varying ability to cause disease. *PLoS One* 2014;9:e112967.
 35. Jalaluddin S, De Mol P, Hemelhof W, Bauma N, Brasseur D, Hennart P, et al. Isolation and characterization of enteroaggregative *Escherichia coli* (EAggEC) by genotypic and phenotypic markers, isolated from diarrheal children in Congo. *Clin Microbiol Infect* 1998;4:213-9.
 36. Chan KN, Phillips AD, Knutton S, Smith HR, Walker-Smith JA. Enteroggregative *Escherichia coli*: another cause of acute and chronic diarrhoea in England? *J Pediatr Gastroenterol Nutr* 1994;18:87-91.
 37. Cohen MB, Nataro JP, Bernstein DI, Hawkins J, Roberts N, Staat MA. Prevalence of diarrheagenic *Escherichia coli* in acute childhood enteritis: a prospective controlled study. *J Pediatr* 2005;146:54-61.
 38. Boisen N, Scheutz F, Rasko DA, Redman JC, Persson S, Simon J, et al. Genomic Characterization of enteroaggregative *Escherichia coli* from children in Mali. *J Infect Dis* 2012;205:431-44.
 39. Huppertz HI, Rutkowski S, Aleksic S, Karch H. Acute and chronic diarrhoea and abdominal colic associated with enteroaggregative *Escherichia coli* in young children living in western Europe. *Lancet* 1997;349:1660-2.
 40. Usein CR, Tatu-Chitoiu D, Ciontea S, Condei M, Damian M. *Escherichia coli* pathotypes associated with diarrhea in Romanian children younger than 5 years of age. *Jpn J Infect Dis* 2009;62:289-93.
 41. Nair GB, Ramamurthy T, Bhattacharya MK, Krishnan T, Ganguly S, Saha DR, et al. Emerging trends in the etiology of enteric pathogens as evidenced from an active surveillance of hospitalized diarrhoeal patients in Kolkata, India. *Gut Pathog* 2010;2:4.
 42. Bangar R, Mamatha B. Identification of enteroaggregative *Escherichia coli* in infants with acute diarrhea based on biofilm production in Manipal, south India. *Indian J Med Sci* 2008;62:8-12.
 43. Clinical Laboratory and Standards Institute. Performance Standards for Antimicrobial Disk and Dilution Susceptibility Tests for Bacteria Isolated From Animals; Approved Standard. VET01-A4. Clin Lab Stand Inst - Fourth Ed. 2013. https://clsi.org/media/1531/vet01a4_sample.pdf
 44. Hassan A, Usman J, Kaleem F, Omair M, Khalid A, Iqbal M. Evaluation of different detection methods of biofilm formation in the clinical isolates. *Brazilian J Infect Dis* 2011;15:305-11.
 45. Polotsky Y, Nataro JP, Kotler D, Barrett TJ, Orenstein JM. HEp-2 cell adherence patterns, serotyping, and DNA analysis of *Escherichia coli* isolates from eight patients with AIDS and chronic diarrhea. *J Clin Microbiol* 1997;35:1952-8.
 46. Gupta D, Sharma M, Sarkar S, Thapa BR, Chakraborti A. Virulence determinants in enteroaggregative *Escherichia coli* from north India and their interaction in vitro organ culture system. *FEMS Microbiol Lett* 2016;363:fnw189.
 47. Meng CY, Smith BL, Bodhidatta L, Richard SA, Vansith K, Thy B, et al. Etiology of diarrhea in young children and patterns of antibiotic resistance in Cambodia. *Pediatr Infect Dis J* 2011;30:331-5.
 48. Nataro JP, Kaper JB, Robins-Browne R, Prado V, Vial P, Levine MM. Patterns of adherence of diarrheagenic *Escherichia coli* to HEp-2 cells. *Pediatr Infect Dis* 1987;6:829-31.
 49. Schmidt H, Knop C, Franke S, Aleksic S, Heesemann J, Karch H. Development of PCR for screening of enteroaggregative *Escherichia coli*. *J Clin Microbiol* 1995;33:701-5.
 50. Albert MJ, Faruque ASG, Faruque SM, Sack RB, Mahalanabis D. Case-control study of enteropathogens associated with childhood diarrhea in Dhaka, Bangladesh. *J Clin Microbiol* 1999;37:3458-64.
 51. Nataro JP, Mai V, Johnson J, Blackwelder WC, Heimer R, Tirrell S, et al. Diarrheagenic *Escherichia coli* infection in Baltimore, Maryland, and New Haven, Connecticut. *Clin Infect Dis* 2006;43:402-7.
 52. Cennimo D, Abbas A, Huang DB, Chiang T. The prevalence and virulence characteristics of enteroaggregative *Escherichia coli* at an urgent-care clinic in the USA: a case-control study. *J Med Microbiol* 2009;58:403-7.
 53. Bolick DT, Roche JK, Hontecillas R, Bassaganya-Riera J, Nataro JP, Guerrant RL. Enteroggregative *Escherichia coli* strain in a novel weaned mouse model: Exacerbation by malnutrition, biofilm as a virulence factor and treatment by nitazoxanide. *J Med Microbiol* 2013;62:896-905.
 54. Dutta S, Guin S, Ghosh S, Pazhani GP, Rajendran K, Bhattacharya MK, et al. Trends in the prevalence of diarrheagenic *Escherichia coli* among hospitalized diarrheal patients in Kolkata, India. *PLoS One* 2013;8:e56068.
 55. Zaidi MB, Campos FD, Estrada-García T, Gutierrez F, León M, Chim R, et al. Burden and transmission of zoonotic foodborne disease in a rural community in Mexico. *Clin Infect Dis* 2012;55:51-60.
 56. Ochoa TJ, Ecker L, Barletta F, Mispireta ML, Gil AI, Contreras C, et al. Age-related susceptibility to infection with diarrheagenic *E. coli* in infants from peri-urban areas of Lima, Peru. *Clin Infect Dis* 2009;49:1694-702.
 57. Nüesch-Inderbinen MT, Hofer E, Hächler H, Beutin L, Stephan R. Characteristics of enteroaggregative *Escherichia coli* isolated from healthy carriers and from patients with diarrhoea. *J Med Microbiol* 2013;62(Pt 12):1828-34.
 58. Chattaway MA, Dallman T, Okeke IN, Wain J. Enteroggregative *E. coli* O104 from an outbreak of HUS in Germany 2011, could it happen again? *J Infect Dev Ctries* 2011;5:425-36.
 59. Lima IFN, Boisen N, Da Silva Quetz J, Havt A, De Carvalho EB, Soares AM, et al. Prevalence of enteroaggregative *Escherichia coli* and its virulence-related genes in a case-control study among children from north-eastern Brazil. *J Med Microbiol* 2013;62:683-93.
 60. Regua-Mangia AH, Gomes TAT, Vieira MAM, Irino K, Teixeira LM. Molecular typing and virulence of enteroaggregative *Escherichia coli* strains isolated from children with and without diarrhoea in Rio de Janeiro city, Brazil. *J Med Microbiol* 2009;58:414-22.
 61. Kosek M, Haque R, Lima A, Babji S, Shrestha S, Qureshi S, et al. Fecal markers of intestinal inflammation and permeability associated with the subsequent acquisition of linear growth deficits in infants. *Am J Trop Med Hyg* 2013;88:390-6.
 62. Okeke IN, Ojo O, Lamikanra A, Kaper JB. Etiology of acute diarrhea in adults in Southwestern Nigeria. *J Clin Microbiol* 2003;41:4525-30.
 63. Boisen N, Østerlund MT, Joensen KG, Santiago AE, Mandomando I, Cravioto A, et al. Redefining enteroaggregative *Escherichia coli* (EaEC): genomic characterization of epidemiological eaec strains. *PLoS Negl Trop Dis* 2020;14:e0008613.
 64. Morin N, Santiago AE, Ernst RK, Guillot SJ, Nataro JP. Characterization of the AggR regulon in enteroaggregative *Escherichia coli*. *Infect Immun* 2013;81:122-32.
 65. Morin N, Santiago AE, Ernst RK, Guillot SJ, Nataro JP. Characterization of the AggR regulon in enteroaggregative *Escherichia coli*. *Infect Immun* 2013;81:122-32.
 66. Amin M, Sirous M, Javaherizadeh H, Motamedifar M, Saki M, Veisi H, et al. Antibiotic resistance pattern and molecular characterization of extended-spectrum β -lactamase producing enteroaggregative *Escherichia coli* isolates in children from southwest Iran. *Infect Drug Resist* 2018;11:1097-104.
 67. Aslani MM, Alikhani MY, Zavari A, Yousefi R, Zamani AR. Characterization of enteroaggregative *Escherichia coli* (EAEC) clinical isolates and their antibiotic resistance pattern. *Int J Infect Dis* 2011;15:e136-9.
 68. Aslani MM, Alikhani MY, Zavari A, Yousefi R, Zamani AR. Characterization of enteroaggregative *Escherichia coli* (EAEC) clinical isolates and their antibiotic resistance pattern. *Int J Infect Dis* 2011;15:e136-9.
 69. Sobieszkańska B, Kowalska-Krochmal B, Mowszet K, Pytrus T. Susceptibility to antimicrobial agents of enteroaggregative *Escherichia coli* strains isolated from children with diarrhea. *Przegl Epidemiol* 2003;57:499-503.
 70. Czczulin JR, Whittam TS, Henderson IR, Navarro-Garcia F, Nataro JP. Phylogenetic analysis of enteroaggregative and diffusely adherent *Escherichia coli*. *Infect Immun* 1999;67:2692-9.
 71. Dallman TJ, Chattaway MA, Cowley LA, Doumith M, Tewolde R, Wooldridge DJ, et al. An investigation of the diversity of strains of enteroaggregative *Escherichia coli* isolated from cases associated with a large multi-pathogen foodborne outbreak in the UK. *PLoS One* 2014;9:e8103.

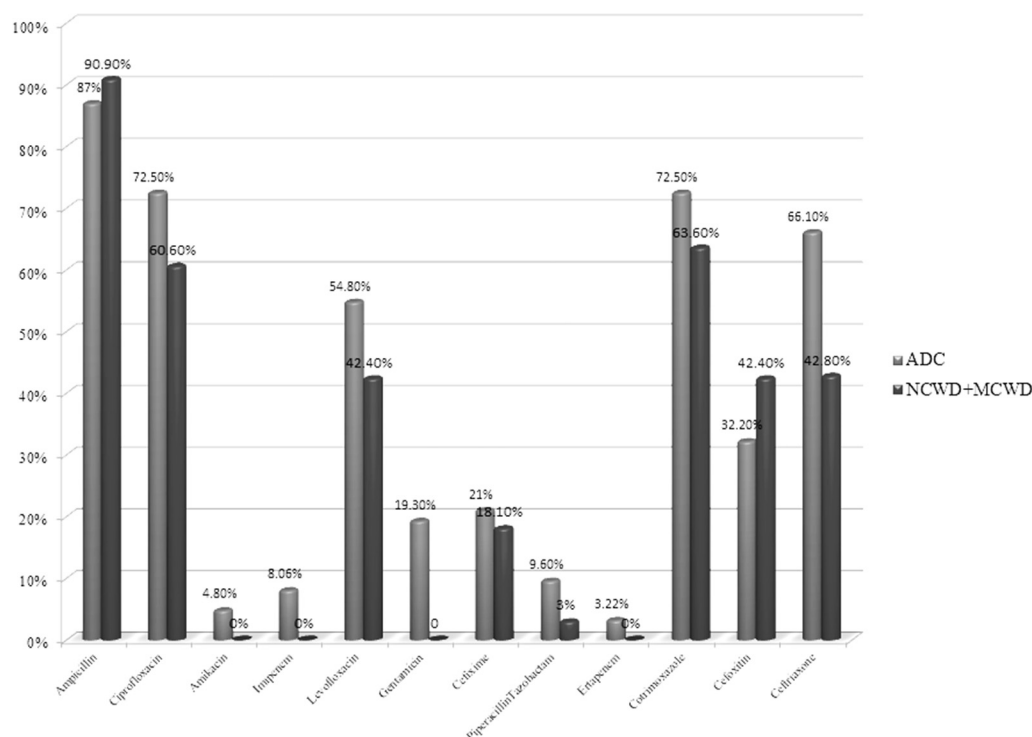


Figure 1. Comparative antimicrobial susceptibility of EAEC from children with acute diarrhea and nourished and malnourished nondiarrheal control groups. ADC, children with acute diarrhea; MCWD and NCWD, nourished and malnourished nondiarrheal control groups.

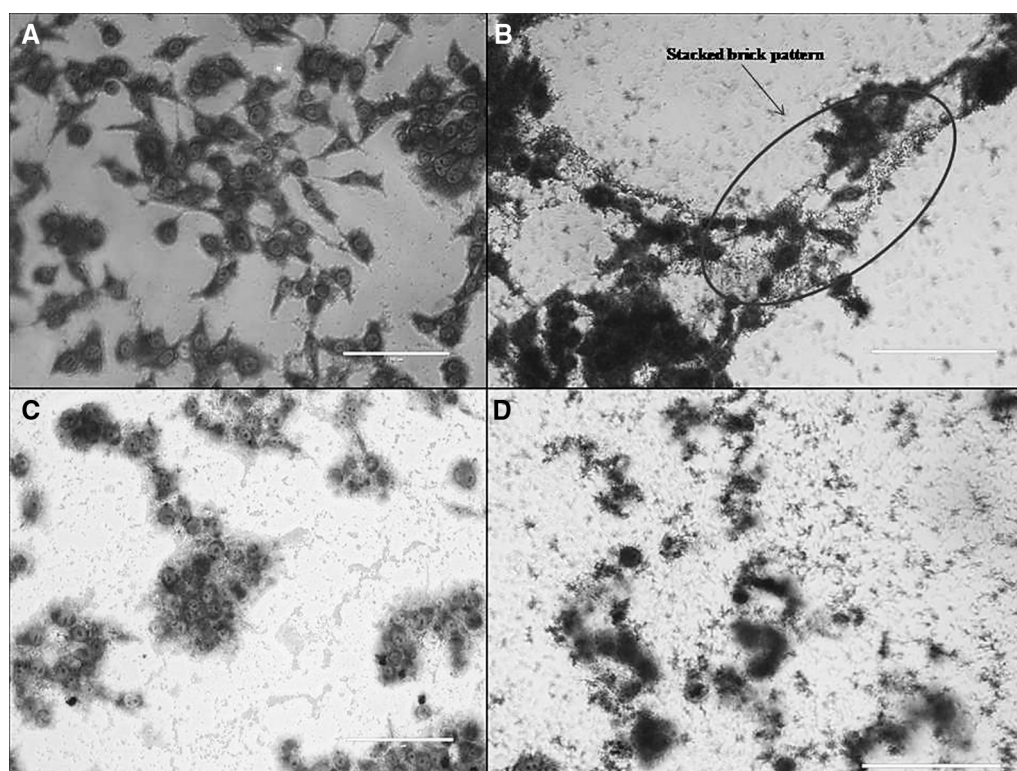


Figure 2. The HEp-2 adhesion assay detected aggregative adherence patterns of EAEC. **A**, Control. **B**, Type A: typical aggregative adherence pattern with honeycomb formation. **C**, Type B: aggregative adherence pattern without typical honeycomb formation. **D**, Type C: aggregative adherence pattern with lined-up cells.

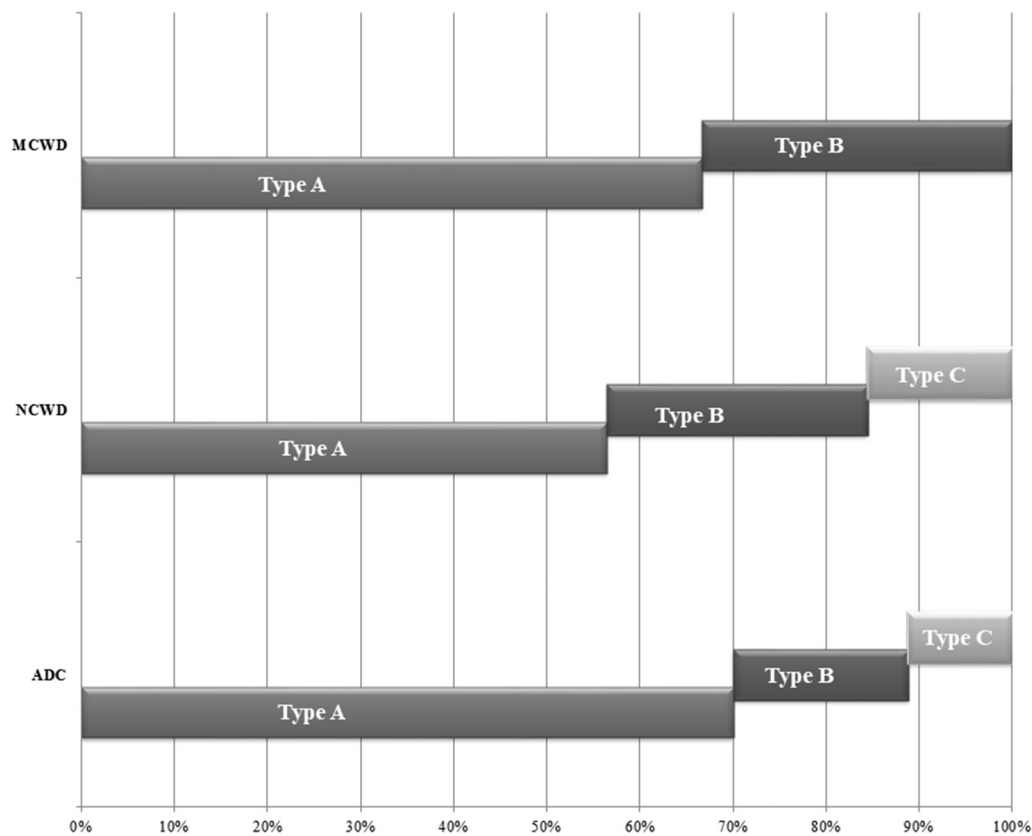


Figure 3. Incidence of each aggregative adherence pattern in EAEC from children with acute diarrhea and nourished and malnourished nondiarrheal control groups.

Table I. List of the target gene, primers sequences, and amplified PCR product size for detection of DEC pathotypes

Target genes	Primer sequence	PCR product size, bp	Primer designation	Reference
<i>bfp</i>	GGAAGTCAAATTCATGGGGGTAT GGAATCAGACGCAGACTGGTAGT	300	Bfp	34
<i>eae</i>	TCAATGCAGTTCGGTTATCAGTT GTAAAGTCCGTTACCCCAACCTG	482	Eae	34
<i>elt</i>	ACGGCGTTACTATCCTCTC TGGTCTCGGTCAGATATGTG	273	LT	35
<i>CVD432</i>	CTGGCGAAAGACTGTATCAT AATGTATAGAAATCCGCTGTT	630	pCVD432	36
<i>estA1</i>	TCTTTCCCTCTTTTAGTCAG ACAGGCAGGATTACAACAAAG	166	STp	35
<i>estA2-4</i>	TTCACCTTTCCCTCAGGATG CTATTCATGCTTTCAGGACCA	120	STh	35
<i>stx1</i>	CAGTTAATGTGGTGGCGAAGG CACCAGACAATGTAACCGCTG	348	Stx1	34
<i>stx2</i>	ATCCTATTCCCGGGAGTTTACG GCGTCATCGTATACAGGAGC	584	Stx2	34
<i>stx1+stx2</i>	GAGCGAAATAATTTATATGTG TGATGATGGCAATTCAGTAT	518	VTcom	37

LT, heat-labile; STh, heat-stable (human); STp, heat-stable (porcine).

Table II. Primers with description of the target gene, product size in base pairs, and annealing temperature, used for the 4 M-PCRs and 3 monoplex PCRs

Target gene and descriptions	Type of PCR	Primer sequence (5' - 3')	PCR product, size, bp	GenBank accession no.	Annealing temperature, °C	References
<i>astA</i> : EAST-1 heat-stable toxin	M-PCR 1	ATGCCATCAACACAGTAT	110	L11241	58	39
<i>Pet</i> : Plasmid-encoded toxin		GCGAGTGACGGCTTTGTAGT				
		GGCACAGAATAAAGGGGTGTTT	302	AF056581	58	40
		CCTCTTGTTCCACGACATAC				
<i>sigA</i> : IgA protease-like homolog		CCGACTTCTCACTTTCTCCCG	430	NC_004337	58	38
		CCATCCAGCTGCATAGTGTG				
<i>pic</i> : Serine protease precursor		ACTGGATCTTAAGGCTCAGGAT	572	AF097644	58	40
		GACTTAATGTCACTGTTCAAGC				
<i>sepA</i> : Shigella extracellular protease		GCAAGTGGAAATATGATCGCG	794	Z48219	58	40
		TTGTTCAAGTCGGAGAAGAAGC				
<i>sat</i> : Secreted autotransporter toxin		TCAGAAGCTCAGCGAATCATTG	932	AE014075	58	38
		CCATTATCACCAGTAAACGCACC				
ORF3: Cryptic protein	M-PCR 2	CAGCAACCATCGCATTTCTA	121	AB261016.2	57	38
		CGCATCTTTCAATACCTCCA				
<i>aap</i> : Dispersin, antiaggregation protein		GGACCCGTCCTCAATGTATAA	250	Z32523	57	38
		CCATTGCGTTAGAGCAGCAT				
<i>aaiC</i> : AaiC, secreted protein		TGGTGACTACTTTGATGGACATTG	313	AB255435.1	57	38
		GACACTCTCTTCTGGGGTAAACGA				
<i>aggR</i> : Transcriptional activator		GCAATCAGATTAARCAAGCGATACA	426	Z18751	57	38
		CATTCTTGATTGCATAAGGATCTGG				
<i>agg4A</i> : AAF/IV fimbrial subunit	M-PCR 3	TGAGTTGTGGGGCTAYCTGGA	169	EU637023	57	38
		CACCATAAGCCGCCAAATAAGC				
<i>aggA</i> : AAF/I fimbrial subunit		TCTATCTRGGGGGGCTAACGCT	220	Y18149	57	38
		ACCTGTTCCTCATAACAGACC				
<i>aafA</i> : AAF/II fimbrial subunit		CTACTTTATTATCAAGTGGAGCCGCTA	289	AY344586	57	38
		GGAGAGGCCAGAGTGAATCCTG				
<i>agg3A</i> : AAF/III fimbrial subunit		CCAGTTATTACAGGGTAACAAGGGAA	370	AF411067	57	38
		TTGGTCTGGAATAACAACCTGAACG				
<i>aafC</i> : Usher, AAF/II assembly unit		ACAGCCTGCGGTCAAAAGC	491	AF114828	57	38
		GCTTACGGGTACGAGTTTACGG				
ORF61: Plasmid-encoded hemolysin	M-PCR 4	AGCTCTGGAACTGGCCTCT	108	Z15104	57	38
		AACCGTCTGATTCTGCTT				
<i>eilA</i> : Salmonella HilA homolog		AGGTCTGGAGCGCGAGTGT	248	CP009685.1	57	38
		GTAAACGGTATCCACGACC				
<i>capU</i> : Hexosyltransferase homolog		CAGGCTGTTGCTCAAATGAA	395	AF134403	57	38
		GTTCCGACATCCTTCTGCTC				
<i>air</i> : Enterotoxigenic immunoglobulin repeat protein		TTATCCTGGTCTGTCTCAAT	600	CP009685.1	57	38
		GGTTAAATCGCTGGTTTCTT				
<i>espY2</i> : Non-LEE-encoded type III secreted effector	Singleplex PCR	CGCAAAAGATCCGGAAAATA	216	ECSP_0073	57	38
		TCAGATTGCTCAGGTCAAC				
<i>rmoA</i> : Putative hemolysin expression-modulating protein	Singleplex PCR	TTACCTTACATATTTCCATATC	210	ECUMN_0072	59	38
		CGAAAACAAAACAGGAATGG				
<i>shiA</i> : shiA-like inflammation suppressor	Singleplex PCR	CAGAATGCCCGCGTAAGGC	292	ECB_03517	57	41
		CACTGAAGGCTCGCTCATGATCGCCG				

Table III. Primers used for the triplex PCRs target gene description, base-pair size, annealing temperature, and primers concentration

Genes/targets	Primer sequence (5' - 3')	PCR product, bp	Annealing temperature primer concentration (°C), pmol/μL	References
ChuA	GACGAACCAACGGTCAGGT TGCCGCCAGTACCAAAGAA	279	55/25	42
YjaA	TGAAGTGTGAGGAGACGCTG GAGTAATGTCGGGGCATTCA	211	55/25	42
TspE4C2	GAGTAATGTCGGGGCATTCA CGCGCCAACAAAGTATTACG	152	52/25	42

Table VII. Distribution of DEC pathotypes among children in all groups

DEC categories	No. of strains (%)		P value
	Children with acute diarrheal, 548 n (%)	Nourished and malnourished controls without diarrhea, 660 n (%)	
Total DEC	166 (30)	92 (14)	.0001*
EAEC	86 (16)	36 (5.4)	.0001*
ETEC	36 (6.5)	30 (4.5)	.12
EPEC	42 (7.6)	25 (3.7)	.003*

*Statistically significant ($P < .05$) when compared pooled prevalence of DEC pathotypes in diarrheal group were compared with nourished and malnourished children in combination. The Fisher exact test was used to compare the presence of DEC in children without diarrhea at different age groups.

Table VIII. Severity of children with EAEC infection in the acute diarrheal group based on the number, age, and sex

Severity	No. of EAEC detected, no. (%)	Sex-wise distribution, male/female	Age-wise distribution, severe/mild/moderate	
Severe	23 (27.7%)	15/8	0-1 y	8/5/3
Mild	34 (40.9%)	20/14	1-2 y	11/9/10
Moderate	26 (31.3%)	16/10	2-5 y	10/19/11

Table IX. Number of the malnourished population in terms of underweight, stunting, wasting

Age groups, y	Underweight		Stunting		Wasting	
	Severe, no. (%)	Moderate, no. (%)	Severe, No. (%)	Moderate, No. (%)	Severe, No. (%)	Moderate, no. (%)
0-1 (n = 25)	9 (8.1)	16 (14.5)	16 (14.5)	9 (8.1)	12 (10.9)	35 (31.8)
1-2 (n = 30)	13 (11.8)	17 (15.4)	16 (14.5)	14 (12.7)	19 (17.2)	11 (10)
2-5 (n = 55)	23 (20.9)	32 (29)	31 (28)	22 (20)	20 (18.1)	13 (11.8)
Total	45 (40.9)	65 (59)	63 (57.2)	45 (40.9)	51 (46.3)	59 (53.6)

Table XI. Different combinations of virulence genes among diarrheal and nondiarrheal stools

Sources	Gene combination	No. (%) of strains
Diarrheal stool	<i>astA</i> , ORF3, <i>aap</i> , <i>aggR</i> , <i>agg4A</i> , ORF61	1 (1.1)
	<i>astA</i> , ORF3, <i>aap</i> , <i>aggR</i> , ORF61	4 (4.6)
	<i>astA</i> , ORF3, <i>aap</i> , <i>aggR</i> , <i>agg4A</i>	4 (4.6)
	<i>astA</i> , <i>aap</i> , <i>aggR</i> , <i>agg4A</i>	6 (6.9)
	<i>astA</i> , ORF3, <i>aap</i> , <i>aggR</i>	9 (10.4)
	<i>astA</i> , ORF3, <i>aggR</i> , <i>agg4A</i>	2 (2.3)
	<i>astA</i> , <i>pet</i> , ORF3, <i>aap</i> , <i>aggR</i> , ORF61	2 (2.3)
	<i>astA</i> , <i>aggR</i> , <i>agg4A</i> , ORF61	2 (2.3)
	<i>astA</i> , <i>aap</i> , <i>aggR</i> , <i>agg4A</i> , ORF61	4 (4.6)
	<i>astA</i> , <i>pet</i> , <i>aap</i> , <i>aggR</i>	2 (2.3)
	<i>astA</i> , <i>aggR</i> , <i>agg4A</i>	10 (11.6)
	<i>astA</i> , <i>aggR</i> , ORF61	3 (3.4)
	<i>astA</i> , <i>aap</i> , <i>aggR</i>	8 (9.3)
	<i>astA</i> , <i>aap</i>	6 (6.9)
	<i>astA</i> , <i>agg4A</i>	2 (2.3)
	<i>aggR</i> , ORF61	4 (4.6)
	<i>astA</i> , <i>aggR</i>	10 (11.6)
	<i>astA</i>	2 (2.3)
	<i>aap</i>	2 (2.3)
	None	2 (2.3)
Nondiarrheal stool	<i>astA</i> , <i>orf3</i> , <i>aap</i> , <i>aggR</i> , <i>agg4A</i> , ORF61	3 (8.3)
	<i>astA</i> , <i>pet</i> , ORF3, <i>aap</i> , <i>aggR</i> , ORF61	1 (2.7)
	<i>astA</i> , <i>pet</i> , ORF3, <i>aap</i> , ORF61	1 (2.7)
	<i>astA</i> , <i>pet</i> , <i>orf3</i> , <i>aggR</i> , <i>agg4A</i> , <i>orf61</i>	1
	<i>astA</i> , ORF3, <i>aggR</i> , ORF61	2 (5.5)
	<i>astA</i> , <i>pet</i> , <i>aap</i> , <i>aggR</i> , ORF61	1 (2.7)
	<i>astA</i> , ORF3, <i>aggR</i> , ORF61	1 (2.7)
	<i>astA</i> , ORF3, <i>aap</i> , <i>aggR</i>	2 (5.5)
	<i>astA</i> , <i>pet</i> , ORF61	1 (2.7)
	<i>AstA</i> , <i>pet</i> , <i>aggR</i> , <i>agg4A</i>	1
	<i>astA</i> , <i>aap</i> ,	2 (2.7)
	<i>astA</i> , <i>pet</i> , <i>aap</i> , <i>agg4A</i>	1 (2.7)
	<i>astA</i> , <i>pet</i> , <i>orf3</i> , ORF61	2 (5.5)
	<i>astA</i> , <i>pet</i> , ORF61	2 (5.5)
	<i>astA</i> , <i>aap</i> , <i>orf61</i>	1
	<i>astA</i> , <i>pet</i> , <i>aap</i> ,	1 (2.7)
	<i>astA</i> , ORF3, <i>aap</i> , ORF61	1 (2.7)
	<i>astA</i> , <i>aggR</i> , <i>agg4A</i> , ORF61	1 (2.7)
	<i>astA</i> , ORF3, ORF61	4 (11)
	<i>astA</i> , <i>aap</i>	4 (11)
	<i>astA</i> , <i>aggR</i>	1 (2.7)
	ORF3, <i>aap</i> , ORF61	1 (2.7)
	ORF3, ORF61	1 (2.7)
	<i>astA</i>	1 (2.7)
	ORF61	1 (2.7)

Table XII. Phylogenetic grouping of EAEC among children in all groups

Phylogroups	Group 1 acute diarrhea N = 86 (%)	Group 2 nourished N = 33 (%)	Group 3 malnourished N = 3 (%)	Total N = 122 (%)	P value
A	6 (6.9)	5 (15.1)	0 (0)	11 (9.0)	.2
B1	22 (25.5)	13 (39.3)	2 (66.6)	37 (30.3)	.19
B2	26 (30.2)	4 (12.1)	1 (33.3)	31 (25.4)	.06
D	34 (37.2)	11 (33.3)	0 (0)	43 (35.2)	.5

*Statistically significant ($P < .05$) when EAEC strains from the acute diarrheal group were compared with nourished and malnourished control groups. Data was analyzed by using the Fisher exact test.